

Woosang Kim

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Education

The University of Electro-Communications

Apr 2024 - Mar 2026

Master of Engineering, Department of Informatics

Advisor: Takuya Nojima

The University of Electro-Communications

Apr 2020 - Mar 2024

Bachelor's degree, Informatics and Computer Engineering

Thesis: Study on a Method for Stimulation of Periodontal Ligament Receptors by Transcutaneous Electrical Stimulation

Advisor: Takuya Nojima

Research Interests

Intraoral haptics; electrotactile stimulation; food texture perception; human-computer interaction; interactive interfaces.

Publications and Presentations

Conference Poster

Characterizing Gingival Electrical Stimulation Toward Intraoral Texture Display

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima.

IEEE VR 2026. Poster. | [Official program](#)

Submitted Manuscript

An Exploratory Study of Chewing-Synchronized Intraoral Electrotactile Stimulation for Modulating Texture-Related Perception During Mastication

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima.

Submitted to *IEEE Transactions on Haptics*, 2026.

Domestic Conference Presentations

Investigation of Oral Sensory Information for Selective Stimulation of Periodontal Ligament Receptors

Woosang Kim, Shiori Iwahama, Sho Sakurai, Koichi Hirota, and Takuya Nojima.

The Virtual Reality Society of Japan, Tokyo, Japan, 2023.

Investigation of Transcutaneous Electrical Stimulation to Periodontal Ligament Receptors and the Oral Sensory Information Generated

Woosang Kim, Sho Sakurai, Koichi Hirota, and Takuya Nojima.

The Virtual Reality Society of Japan, SIG Haptics, Miyagi, Japan, 2023.

Research Experience

Nojima Laboratory, UEC

Apr 2024 - Mar 2026

Graduate Researcher | Advisor: Takuya Nojima

Investigated intraoral electrical stimulation and texture-related perception during mastication. This work includes the IEEE VR 2026 poster and the manuscript submitted to IEEE Transactions on Haptics.

University of Chicago

Jul - Aug 2024

Summer Research Intern, Human Computer Integration Lab | Advisor: Pedro Lopes

Developed and tested electrical stimulation waveforms to explore sensory experiences related to the trigeminal nerve.

Nojima Laboratory, UEC

Apr 2023 - Mar 2024

Undergraduate Researcher | Advisor: Takuya Nojima

Investigated stimulation methods targeting periodontal ligament receptors. Designed electrode placement with Fusion 360 and 3D printing, developed Arduino and Processing prototypes, and led experiment design and execution.

Industry Experience

NTT DOCOMO

Apr 2026 - Present

Research and Development Engineer | Tokyo, Japan

Selected Projects

Mask Interface | Co-leader | 2023

Directed mask and hardware development, contributed to programming and interface design, and coordinated the team. Exhibited at IVRC and U☆PoC; received the Aflac Inc. Award.

Branco | Project management, ethnography, and idea design

Contributed to interviews and a brand concept exploring happiness for the contest jointly organized by Hakuodo and the University of Tokyo.

GakkiMakura | Idea design, hardware, and software

Contributed to a musical pillow combining touch and pressure sensing, sound output, and vibration feedback.

Honors

Aflac Inc. Award, UEC Idea Demonstration Contest (2023)

Hirose Scholarship Foundation (Apr 2023 - Mar 2024)

Hasegawa International Scholarship Foundation (Apr 2021 - Mar 2023)

Activities and Teaching

UT-Virtual, Member (Sep 2020 - Mar 2022). Organized an exchange with the University of Maryland XR Club and developed VR projects for exhibition.

UEC First-year Student Support, Learning Assistant (Apr 2021 - Mar 2022). Helped students adjust to university life and coursework during the COVID-19 pandemic.

Technical Skills and Languages

Programming: C, Processing, Python, MATLAB

Tools: Arduino, Fusion 360, Unity, GitHub

Languages: Korean (native), Japanese (fluent), English (advanced)